B. Tech 1st Semester Sessional 1

Sub: Mathematics I, Total Marks-20, Branch-Civil Engineeri

Date: 29.09.26

nswer **(any four)** of the following questions: 5X4=20

1. Expand $f(x) = \log(x+1)$ in powers of x up to three nonzero terms
2. Expand the function $f(x) = x^4 + 3x^3 - 2x + 3$ in powers of $x - 1$
3. Evaluate: (i) $\lim_{x\to 0} \frac{x-\sin x}{x}$ (ii) $\lim_{x\to 2} \frac{x^2-4}{x-2}$
4. If $f(x,y) = \sin(x^2 + xy + y^2)$, evaluate $\frac{\partial^2 f}{\partial x^2}$ and $\frac{\partial^2 f}{\partial y^2}$
5. If $z = x^2 - xy$, where $x = t^2, y = e^{-2t}$, then find $\frac{dz}{dt}$
6. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{3}{(x+y+z)^2}$
7. If $u = \frac{x^3-y^3}{x+y}$, then apply Euler's theorem to evaluate $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$